The Laplace–Beltrami Operator in Spherical and Geodesic Normal Coordinates

Contents

Let (M,g) be an N-dimensional Riemannian manifold. In a local coordinate system $(\xi^1,\dots,\xi^N)$, the Laplace–Beltrami operator acting on a smooth function f is

$$\Delta_g f = \frac{1}{\sqrt{\det g}}\frac{\partial}{\partial \xi^i}\left(\sqrt{\det g}\, g^{ij}\frac{\partial f}{\partial \xi^j}\right),$$

where $g=(g_{ij})$, $(g^{ij})=(g_{ij})^{-1}$, and the Einstein summation convention is understood.

It is often useful to write the same expression in the expanded form

$$\Delta_g f = g^{ij}\frac{\partial^2 f}{\partial \xi^i \partial \xi^j} + \left(\frac{\partial g^{ij}}{\partial \xi^i} + g^{ij}\frac{\partial}{\partial \xi^i}\log\sqrt{\det g}\right)\frac{\partial f}{\partial \xi^j}. \tag{1}$$

Thus the quantity $\log\sqrt{\det g}$ records precisely the contribution of the Riemannian volume distortion to the first-order part of the Laplacian.

We first recover the familiar Euclidean formula in ordinary spherical coordinates and then describe its natural analogue in geodesic normal coordinates on a general Riemannian manifold. Standard background on normal coordinates, geodesic polar coordinates, and curvature expansions may be found in [1, 2, 3].

1 Euclidean Space in Spherical Coordinates

Consider $\mathbb{R}^N$ with its Euclidean metric and introduce spherical coordinates $(r,\theta^1,\dots,\theta^{N-1})$, where $r=|x|$ and $\theta=(\theta^1,\dots,\theta^{N-1})\in\mathbb{S}^{N-1}$.

Let $g_{\mathbb{S}^{N-1}}=(g^{\mathbb{S}^{N-1}}_{ab})$ denote the standard metric on the unit sphere.

1.1 Metric tensor

The Euclidean metric decomposes as $\mathrm{d}s^2=\mathrm{d}r^2+r^2\,\mathrm{d}s^2_{\mathbb{S}^{N-1}}$. Equivalently,

$$g=\begin{pmatrix}1 & 0\\ 0 & r^2 g^{\mathbb{S}^{N-1}}_{ab}(\theta)\end{pmatrix}.$$

Consequently, $g^{rr}=1$, $g^{ra}=0$, and $g^{ab}=r^{-2}g^{ab}_{\mathbb{S}^{N-1}}$.

Since the angular block has dimension $N-1$, we have $\det g = r^{2(N-1)}\det g_{\mathbb{S}^{N-1}}$ and $\sqrt{\det g}=r^{N-1}\sqrt{\det g_{\mathbb{S}^{N-1}}}$. Thus $\log\sqrt{\det g}=(N-1)\log r+\log\sqrt{\det g_{\mathbb{S}^{N-1}}}$, so $\partial_r\log\sqrt{\det g}=(N-1)/r$.

1.2 Radial contribution

The radial part of the Laplace–Beltrami operator is

$$\begin{aligned}\frac{1}{\sqrt{\det g}}\frac{\partial}{\partial r}\left(\sqrt{\det g}\,g^{rr}\frac{\partial f}{\partial r}\right) &= \frac{1}{r^{N-1}\sqrt{\det g_{\mathbb{S}^{N-1}}}}\frac{\partial}{\partial r}\left(r^{N-1}\sqrt{\det g_{\mathbb{S}^{N-1}}}\frac{\partial f}{\partial r}\right)\\ &= \frac{1}{r^{N-1}}\frac{\partial}{\partial r}\left(r^{N-1}\frac{\partial f}{\partial r}\right).\end{aligned}$$

Expanding the derivative gives $\partial_r^2 f+\frac{N-1}{r}\partial_r f$. The coefficient of the first radial derivative is exactly $\partial_r\log\sqrt{\det g}=(N-1)/r$.

1.3 Angular contribution

The angular part is

$$\begin{aligned}&\frac{1}{\sqrt{\det g}}\frac{\partial}{\partial \theta^a}\left(\sqrt{\det g}\,g^{ab}\frac{\partial f}{\partial \theta^b}\right)\\ &\quad= \frac{1}{r^{N-1}\sqrt{\det g_{\mathbb{S}^{N-1}}}}\frac{\partial}{\partial \theta^a}\left(r^{N-1}\sqrt{\det g_{\mathbb{S}^{N-1}}}\frac{1}{r^2}g^{ab}_{\mathbb{S}^{N-1}}\frac{\partial f}{\partial \theta^b}\right).\end{aligned}$$

Since r is independent of the angular variables,

$$= \frac{1}{r^2} \frac{1}{\sqrt{\det g_{\mathbb{S}^{N-1}}}} \frac{\partial}{\partial \theta^a} \left(\sqrt{\det g_{\mathbb{S}^{N-1}}} g^{ab}_{\mathbb{S}^{N-1}} \frac{\partial f}{\partial \theta^b} \right).$$

The expression in parentheses is precisely the Laplace–Beltrami operator on the unit sphere. Thus angular part $= \frac{1}{r^2} \Delta_{\mathbb{S}^{N-1}} f$.

1.4 Euclidean spherical-coordinate formula

Combining the radial and angular terms yields

$$\Delta_{\mathbb{R}^N} = \frac{1}{r^{N-1}} \frac{\partial}{\partial r} \left(r^{N-1} \frac{\partial}{\partial r} \right) + \frac{1}{r^2} \Delta_{\mathbb{S}^{N-1}} = \frac{\partial^2}{\partial r^2} + \frac{N-1}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \Delta_{\mathbb{S}^{N-1}}. \tag{2}$$

This formula is the flat-space prototype for the corresponding decomposition in geodesic polar coordinates on a Riemannian manifold.

2 The Laplacian in General Local Coordinates

Before passing to normal coordinates, it is useful to isolate the role of the Riemannian volume density.

Set $\rho(\xi) = \sqrt{\det g(\xi)}$. Then $\Delta_g f = \rho^{-1} \partial_i (\rho\, g^{ij} \partial_j f)$. Expanding the derivative,

$$\begin{aligned} \Delta_g f &= g^{ij} \partial_i \partial_j f + \frac{1}{\rho} \partial_i (\rho g^{ij}) \partial_j f \\ &= g^{ij} \partial_i \partial_j f + \left[\partial_i g^{ij} + g^{ij} \frac{\partial_i \rho}{\rho} \right] \partial_j f. \end{aligned}$$

Since $\partial_i \rho / \rho = \partial_i \log \rho$, this is precisely the expanded expression in (1). This form makes transparent the distinction between two geometric effects:

1. the variation of the inverse metric g^{ij};
2. the variation of the Riemannian volume density $\sqrt{\det g}$.

The second effect is measured directly by $\mathrm{d} \log \sqrt{\det g}$.

3 Riemannian Normal Coordinates

Let (M, g) be an N-dimensional Riemannian manifold and fix a point $p \in M$. Choose an orthonormal basis $e_1, \ldots, e_N$ of T_pM. The exponential map $\exp_p : T_pM \longrightarrow M$ defines Riemannian normal coordinates $(x^1, \ldots, x^N)$ near p through $x = x^i e_i$, $q = \exp_p(x)$.

In these coordinates the center p corresponds to $x = 0$.

3.1 Defining properties at the center

Normal coordinates satisfy $g_{ij}(0) = \delta_{ij}$ and $\partial_k g_{ij}(0) = 0$. Consequently, $g^{ij}(0) = \delta^{ij}$ and $\partial_k g^{ij}(0) = 0$. Moreover, $\det g(0) = 1$ and $\partial_k \log \sqrt{\det g}(0) = 0$, so the first-order coefficients of the Laplace–Beltrami operator vanish at the center.

It follows immediately from (1) that $(\Delta_g f)(p) = \sum_{i=1}^N \partial^2_{x^i} f(0)$.

Thus, at the center of a normal coordinate system, the Laplace–Beltrami operator agrees with the ordinary Euclidean Laplacian.

This statement is pointwise. Away from the center, curvature causes both the metric coefficients and the Riemannian volume density to depart from their Euclidean values.

3.2 Expansion of the volume density

The first nontrivial correction to the Euclidean volume density is controlled by the Ricci tensor. With the standard curvature convention,

$$\begin{aligned}
\sqrt{\det g(x)} &= 1 - \frac{1}{6}\,\mathrm{Ric}_{ij}(p)x^i x^j + O(|x|^3),\\
\log\sqrt{\det g(x)} &= -\frac{1}{6}\,\mathrm{Ric}_{ij}(p)x^i x^j + O(|x|^3),\\
\partial_k \log\sqrt{\det g(x)} &= -\frac{1}{3}\,\mathrm{Ric}_{kj}(p)x^j + O(|x|^2).
\end{aligned}$$

Thus the logarithmic derivative of the volume density vanishes at p but begins to record curvature to first order away from p.

In particular, normal coordinates are Euclidean to first order, whereas curvature appears at second order in the metric and in the volume density.

4 Geodesic Polar Coordinates

Normal coordinates may themselves be decomposed into a radial variable and an angular variable. Write $x = r\theta$, $r = |x|$, $\theta \in \mathbb{S}^{N-1}$. Then $q = \exp_p(r\theta)$.

For sufficiently small $r > 0$, before reaching the cut locus of p, the variables (r, θ) form geodesic polar coordinates.

The radial coordinate is precisely the Riemannian distance from p: $r(q) = d_g(p, q)$.

By the Gauss lemma, the radial direction is orthogonal to the geodesic spheres $S_r(p) = \{q : d_g(p, q) = r\}$. Hence the metric has block form $g = \mathrm{d}r^2 + h_{ab}(r,\theta)\,\mathrm{d}\theta^a\,\mathrm{d}\theta^b$, with no mixed $\mathrm{d}r\,\mathrm{d}\theta^a$ terms.

It is convenient to factor out the Euclidean radial scaling, $h_{ab}(r,\theta) = r^2\gamma_{ab}(r,\theta)$. Thus

$$g = \mathrm{d}r^2 + r^2\gamma_{ab}(r,\theta)\,\mathrm{d}\theta^a\,\mathrm{d}\theta^b. \tag{3}$$

As $r \to 0$, $\gamma_{ab}(r,\theta) \longrightarrow (g_{\mathbb{S}^{N-1}})_{ab}(\theta)$. In Euclidean space this equality holds for every r; on a curved manifold the family $\gamma(r,\cdot)$ varies with r.

4.1 Determinant and volume density

From the block structure, $\det g = \det h_{ab}$. Since $h_{ab} = r^2\gamma_{ab}$ and the angular block has dimension $N-1$, $\det g = r^{2(N-1)}\det\gamma(r,\theta)$, hence $\sqrt{\det g} = r^{N-1}\sqrt{\det\gamma(r,\theta)}$. Taking logarithms gives $\log\sqrt{\det g} = (N-1)\log r + \log\sqrt{\det\gamma(r,\theta)}$, and therefore $\partial_r\log\sqrt{\det g} = (N-1)/r + \partial_r\log\sqrt{\det\gamma(r,\theta)}$.

The first term is the familiar Euclidean contribution. The second term measures the deviation of geodesic spheres from Euclidean spheres and is therefore a direct manifestation of curvature.

5 Laplace–Beltrami Operator in Geodesic Polar Coordinates

We now substitute the geodesic polar form of the metric into the Laplace–Beltrami formula.

Since $g^{rr} = 1$ and $g^{ra} = 0$, while $g^{ab} = r^{-2}\gamma^{ab}(r,\theta)$, the radial part is $\frac{1}{\sqrt{\det g}}\partial_r(\sqrt{\det g}\,\partial_r f)$. Expanding,

$$\begin{aligned}
\frac{1}{\sqrt{\det g}}\frac{\partial}{\partial r}\left(\sqrt{\det g}\frac{\partial f}{\partial r}\right) &= \frac{\partial^2 f}{\partial r^2} + \frac{\partial}{\partial r}\log\sqrt{\det g}\,\frac{\partial f}{\partial r}\\
&= \frac{\partial^2 f}{\partial r^2} + \left[\frac{N-1}{r} + \frac{\partial}{\partial r}\log\sqrt{\det\gamma(r,\theta)}\right]\frac{\partial f}{\partial r}.
\end{aligned}$$

For the angular contribution,

$$\begin{aligned}
&\frac{1}{\sqrt{\det g}}\frac{\partial}{\partial\theta^a}\left(\sqrt{\det g}\,g^{ab}\frac{\partial f}{\partial\theta^b}\right)\\
&\quad= \frac{1}{r^2}\frac{1}{\sqrt{\det\gamma}}\frac{\partial}{\partial\theta^a}\left(\sqrt{\det\gamma}\,\gamma^{ab}\frac{\partial f}{\partial\theta^b}\right).
\end{aligned}$$

For each fixed r, the expression on the right is the Laplace–Beltrami operator associated with the angular metric $\gamma(r,\cdot)$. Denote it by $\Delta_{\gamma(r)}$. We therefore obtain the fundamental decomposition

$$\Delta_g = \frac{\partial^2}{\partial r^2} + \left(\frac{N-1}{r} + \frac{\partial}{\partial r}\log\sqrt{\det\gamma(r,\theta)}\right)\frac{\partial}{\partial r} + \frac{1}{r^2}\Delta_{\gamma(r)}. \tag{4}$$

Equivalently, without factoring $h_{ab} = r^2\gamma_{ab}$, $\Delta_g = \partial_r^2 + (\partial_r \log\sqrt{\det h(r,\theta)})\partial_r + \Delta_{h(r)}$.
This is the Riemannian analogue of the ordinary spherical-coordinate formula.

6 Radial Laplacian and the Geometry of Geodesic Spheres

If $f(q) = F(r(q))$ depends only on the distance from p, then all angular derivatives vanish. Hence $\Delta_g F(r) = F''(r) + (\partial_r \log\sqrt{\det g})F'(r)$. Using the polar volume-density identity above,

$$\Delta_g F(r) = F''(r) + \left[\frac{N-1}{r} + \frac{\partial}{\partial r}\log\sqrt{\det\gamma(r,\theta)}\right]F'(r). \tag{5}$$

In particular, taking $F(r) = r$ gives $\Delta_g r = \partial_r \log\sqrt{\det g} = (N-1)/r + \partial_r \log\sqrt{\det\gamma(r,\theta)}$.

Geometrically, $\Delta_g r$ is the divergence of the radial unit vector field $\nabla r = \frac{\partial}{\partial r}$. Equivalently, up to the standard choice of sign convention for mean curvature, it is the mean curvature of the geodesic sphere $S_r(p)$.

Thus the coefficient of the first radial derivative in the Laplacian has a direct geometric interpretation: it measures the infinitesimal rate of expansion of the area element of geodesic spheres.

7 Curvature Correction Near the Center

The normal-coordinate expansion $\sqrt{\det g(x)} = 1 - \frac{1}{6}\operatorname{Ric}_{ij}(p)x^i x^j + O(|x|^3)$ can be expressed in polar form by setting $x = r\theta$. The Riemannian volume element then has the asymptotic form

$$\mathrm{d}\operatorname{vol}_g = r^{N-1}\left[1 - \frac{1}{6}\operatorname{Ric}_p(\theta,\theta)r^2 + O(r^3)\right]\mathrm{d}r\ \mathrm{d}\sigma(\theta), \tag{6}$$

where $\mathrm{d}\sigma$ is the standard volume element on $\mathbb{S}^{N-1}$.

Equivalently,

$$\sqrt{\det\gamma(r,\theta)} = \sqrt{\det g_{\mathbb{S}^{N-1}}(\theta)}\left[1 - \frac{1}{6}\operatorname{Ric}_p(\theta,\theta)r^2 + O(r^3)\right],$$
$$\log\sqrt{\det\gamma(r,\theta)} = \log\sqrt{\det g_{\mathbb{S}^{N-1}}(\theta)} - \frac{1}{6}\operatorname{Ric}_p(\theta,\theta)r^2 + O(r^3).$$

Hence $\partial_r \log\sqrt{\det g} = (N-1)/r - \frac{1}{3}\operatorname{Ric}_p(\theta,\theta)r + O(r^2)$, and therefore $\Delta_g r = (N-1)/r - \frac{1}{3}\operatorname{Ric}_p(\theta,\theta)r + O(r^2)$.

For a radial function $F(r)$ this gives

$$\Delta_g F(r) = F''(r) + \left[\frac{N-1}{r} - \frac{1}{3}\operatorname{Ric}_p(\theta,\theta)r + O(r^2)\right]F'(r). \tag{7}$$

The Euclidean expression $F''(r) + \frac{N-1}{r}F'(r)$ is therefore the leading term, while the first correction is determined by the Ricci curvature in the radial direction.

8 Euclidean Space as the Flat Special Case

In Euclidean space, $\gamma(r,\theta) = g_{\mathbb{S}^{N-1}}(\theta)$ is independent of r. Hence $\partial_r \log\sqrt{\det\gamma(r,\theta)} = 0$ and $\Delta_{\gamma(r)} = \Delta_{\mathbb{S}^{N-1}}$. The general geodesic-polar formula therefore reduces immediately to the Euclidean spherical-coordinate formula derived in Section 1.

Accordingly, the additional term $\partial_r \log\sqrt{\det\gamma(r,\theta)}$ may be viewed as the correction to the Euclidean radial Laplacian arising from the curvature-induced distortion of the Riemannian volume element.

Remark 1. *The geodesic polar coordinate system is valid only away from the center* $r = 0$ *and, globally, only up to the cut locus of* p*. The apparent singularity* $(N-1)/r$ *is a coordinate singularity analogous to the one in ordinary Euclidean spherical coordinates.*

Remark 2. *For a general Riemannian manifold,* $\mathrm{Ric}_p(\theta, \theta)$ *depends on the angular direction* θ*. Consequently, even when* F *is a function only of the distance* r*, the quantity* $\Delta_g F(r)$ *need not itself be independent of* θ*. Additional symmetry, as occurs for space forms and more generally for harmonic manifolds, is required for the radial Laplacian to depend on* r *alone.*

References

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